





Revinn

EdTech Platform

INTERNAL TECHNICAL DOCUMENT

Bangladesh Bank

Exam AI Platform

Complete Knowledge Base, Syllabus Master Table & Data Strategy

AD · Senior Officer · Officer · Officer (Cash)

MCQ Preparation · Written Preparation · Viva Preparation

June 2026 | Prepared by Revinn Product Team | For CEO & Engineering Review

3 Exam Levels AD · SO · Officer	3 Knowledge Bases MCQ · Written · Viva	7 Core Subjects per level (structured)	5+ Data Sources per pipeline layer	100% Bangla Support all three exam levels
--	---	---	---	--

Table of Contents

1. Bangladesh Bank Exam — Overview & Marks Distribution
2. Syllabus Master Table — Officer (Entry Level)
3. Syllabus Master Table — Senior Officer (Mid Level)
4. Syllabus Master Table — Assistant Director (Advanced Level)
5. Three-Track Comparison — Difficulty & Scope Progression
6. Written Exam Deep-Dive — All Levels
7. Viva Preparation Database
8. Data Collection Strategy — AI Pipeline
9. MCQ Data Sources — Structured View
10. Viva Data Model — Real Exam Behaviour
11. Concept Data — AI Knowledge Brain
12. What Else to Add — Critical Missing Layers
13. Recommended Books & Reading Materials
14. Implementation Roadmap

1. Bangladesh Bank Exam — Overview & Marks Distribution

The Bangladesh Bank (BB) and Bankers' Selection Committee Secretariat (BSCS) conduct examinations for three primary officer-level positions. Each position has a multi-stage selection process: Preliminary MCQ → Written Examination → Viva Voce. The exam difficulty, marks distribution, syllabus depth, and viva expectations differ significantly across the three levels.

1.1 Preliminary MCQ — Marks Distribution by Level

Subject	Officer	Senior Officer	AD (General)	Notes
Bangla (Lit + Grammar)	15	20	25	AD: 8 lit + 17 grammar
English (Grammar + Vocab)	15	20	25	AD: 10 grammar + 15 vocab
Mathematics	15	20	20	Complexity increases per level
General Knowledge	15	20	20	Current affairs weight rises at AD
ICT / Computer	10	10	10	Networking/Fintech deeper at AD
Banking & Finance	—	10	—	SO only in prelim banking section
Reasoning / Analytical	10	—	—	Officer level only in prelim
TOTAL (Prelim)	80	100	100	—
Negative Marking	-0.25	-0.25	-0.25	Per wrong answer, all levels
Duration	45 min	1 hour	1 hour	Timed MCQ test

1.2 Written Exam — Marks Distribution (AD General, Confirmed Official)

Section	Officer	Sr. Officer	AD General	Strategic notes
Comprehension (English passage)	20	25	30	Specific BB answer format required
Mathematics (step-by-step written)	20	25	30	Partial marks per step — format critical
General Knowledge (structured answers)	20	25	30	Statistics + current data expected

Section	Officer	Sr. Officer	AD General	Strategic notes
Translation (Bangla ↔ English)	10	10	10	Register + idiom accuracy scored
Focus Writing — Bangla (রচনা)	20	30	35	Structure + argument depth scored
Focus Writing — English (Essay)	20	30	35	600–1000 words; IELTS-style rubric
Argument Writing (For/Against)	—	—	30	AD-exclusive section; 15% of total
TOTAL (Written)	110	145	200	—
Duration	1.5 hrs	2 hrs	2 hrs	Section time management critical

Key Exam Insight

Focus Writing (Bangla 35 + English 35 = 70 marks) is the single largest component of the AD Written exam — 35% of total written marks.

Argument Writing (30 marks) is AD-exclusive and the most difficult section — no BD platform currently prepares candidates for this specifically.

Negative marking (-0.25 per wrong) in the Preliminary means candidates scoring 68% accuracy should attempt ~85 questions, not all 100.

Viva carries 25 marks for AD and weighs heavily against total score — a 10-mark viva difference can change final selection outcome.

2

Section 2 — Officer (Entry Level)

Syllabus Master Table

ENTRY LEVEL | Bangla · English · Math · GK · ICT · Reasoning · Writing · Viva

Section	Subject	Topics Covered
MCQ	Bangla	Basic grammar · বর্ণ-শব্দ-বাক্য · সমাস (basic) · সন্ধি (basic) · কারক ও বিভক্তি · বাচ্য ও উক্তি পরিবর্তন · এক কথায় প্রকাশ · বাগধারা ও প্রবাদ-প্রবচন · বানান শুদ্ধি · সাহিত্য (basic overview) · গুরুত্বপূর্ণ লেখক ও রচনা
MCQ	English	Basic grammar · Tense · Subject-Verb Agreement · Voice · Narration · Preposition · Vocabulary · Synonyms/Antonyms · Idioms & Phrases · Sentence correction · Fill in the blanks · Basic comprehension
MCQ	Mathematics	Percentage · Ratio & Proportion · Profit & Loss · Simple & Compound Interest · Time & Work · Time & Distance · Basic Algebra
MCQ	General Knowledge	Bangladesh affairs · Liberation War (basic) · Geography of Bangladesh · Current affairs (basic) · UN / WHO / IMF key facts
MCQ	ICT	Computer fundamentals · Hardware vs. software · MS Office (Word, Excel, PowerPoint) · Internet & email · Basic networking concepts
MCQ	Reasoning	Number series · Pattern recognition · Basic logical reasoning
Written	Writing	Short essay · Paragraph writing · Basic translation (Bangla ↔ English) · Simple math steps with working
Viva	Oral	Personal introduction · Career motivation · Basic banking awareness (deposits, loans, central bank role)

Officer Level — Key Characteristics

- Entry point to banking career — largest applicant pool per cycle
- Math is SSC-level — focus on speed and accuracy over complexity
- Viva is brief and personal — 'why banking?' and basic awareness questions
- No Banking & Finance section in MCQ at this level
- Officer (Cash) variant: heavier numerical aptitude, cash management procedures

3

Section 3 — Senior Officer (Mid Level)

Syllabus Master Table

MID LEVEL | 7 MCQ subjects + Advanced Written + Banking Viva

Section	Subject	Topics Covered
MCQ	Bangla	Advanced grammar · Complex sentence correction · সাহিত্য বিশ্লেষণ · ভাষা আন্দোলন ও সাহিত্য ইতিহাস · ছন্দ ও অলংকার
MCQ	English	Cloze test · Error detection · Sentence rearrangement · Long comprehension · Advanced vocabulary · Analogy · Critical reading
MCQ	Mathematics	Algebra · Geometry · Trigonometry basics · Probability & Permutation-Combination · Statistics · Data interpretation
MCQ	General Knowledge	Bangladesh economy (advanced) · Current affairs (national + global) · International politics · Global organizations (IMF, WB, ADB, WTO) · Science & technology
MCQ	ICT	Networking (IP, DNS, HTTP, TCP/IP model) · SQL basics · Cybersecurity basics · Cloud computing concepts · Fintech fundamentals
MCQ	Banking & Finance	Monetary policy · Inflation & deflation · Repo / reverse repo / CRR / SLR · Banking regulations · Financial institutions in BD · Basel I–III basics · Digital banking & MFS
MCQ	Reasoning	Puzzles · Logical reasoning chains · Data interpretation sets · Matrix reasoning
Written	Writing	Essay (400–600 words) · Advanced translation (both directions) · Case-based banking answer · Report writing · Précis writing
Viva	Oral	Banking system operations · Monetary policy basics · ICT and finance intersections · Current national and international affairs

Senior Officer Level — Key Characteristics

- Banking & Finance first appears as an MCQ subject at this level — 10 marks in prelim
- Math escalates to probability, permutation-combination, and data interpretation
- No Argument Writing in Written exam — but Report Writing is expected
- Viva tests operational banking knowledge — can you explain Repo rate clearly?
- Combined Bank exams (11 banks) mostly recruit at Senior Officer level — largest vacancy count

4

Section 4 — Assistant Director / AD General (Advanced)

Syllabus Master Table — Official BB AD Pattern

ADVANCED LEVEL | Prelim 100M · Written 200M · Viva 25M | AD-exclusive: Argument Writing (30M)

Section	Subject	Topics Covered
MCQ	Bangla	Advanced grammar analysis · Literary criticism · Language history (Bangla language evolution) · Classical & modern literature · Chhandas (metre) · পদ প্রকরণ, ধ্বনি পরিবর্তন, লিঙ্গ, বচন
MCQ	English	GRE-level vocabulary · Critical reading comprehension · Sentence transformation · Tricky error correction · Analogy · Tone & inference questions
MCQ	Mathematics	Advanced algebra · Functions & graphs · Linear equations · Probability (advanced) · Permutation-combination · Data interpretation (complex DI sets) · Analytical reasoning
MCQ	General Knowledge	Bangladesh economic policy · Global economy dynamics · Geopolitics · International finance institutions · SDGs & climate-economy nexus · Bangladesh Bank history & governance
MCQ	ICT	Network architecture · OSI model layers · TCP/IP deep concepts · Advanced DBMS & SQL · Cybersecurity frameworks · Fintech systems · Digital transformation · Cloud & distributed systems · MFS regulation in BD
MCQ	Banking & Economics	Monetary policy (deep) · Fiscal policy · Inflation control mechanisms · Exchange rate systems (fixed, floating, managed float) · Financial markets & capital market · Basel III / CAMELS · Risk management framework · Green banking · Central banking operations (BB) · AML/CFT
MCQ	Reasoning	Complex puzzles · DI sets (multi-table) · Logical chains · Case-based reasoning · Critical thinking questions
Written	Essay	Focus Writing Bangla — 35 marks (রচনা structure: ভূমিকা, মূল অংশ, উপসংহার) · Focus Writing English — 35 marks (600–1000 words, structured argument)
Written	Comprehension	30 marks · English passage · BB-specific answer format · Reference-to-text style · Inference and vocabulary questions
Written	Mathematics	30 marks · Step-by-step solutions mandatory · Partial marks per step · Algebra, Probability, DI, Interest, Geometry
Written	General Knowledge	30 marks · Structured answers · Relevant statistics expected · Current data (GDP, inflation, FX reserve, budget)
Written	Translation	10 marks · Bangla to English and English to Bangla · Register accuracy · Idiom handling · False friends awareness
Written	Argument Writing	30 marks · AD-EXCLUSIVE · For/against structured format · Thesis statement · Counterargument + refutation · Logical conclusion · 400–500 words

Section	Subject	Topics Covered
Viva	Oral	Deep banking & economics · Policy interpretation · BB Act 1972 & Bank Company Act 1991 · BB departments (BRPD, FEPD, CMSD, Research) · Global + national current affairs · Situational judgment · Background-specific questions

AD Level — Key Characteristics

- Argument Writing (30 marks) is AD-exclusive — the section most candidates underestimate and under-prepare
- Focus Writing (35+35=70 marks) is 35% of the entire Written exam — the highest-value section
- ICT at AD level expects network architecture, OSI model, and Fintech regulatory knowledge
- Banking & Economics requires policy-level depth: CAMELS rating, Basel III capital tiers, BB circulars
- Viva panel includes BB senior officials — expect questions on BB Act, departmental functions, and current economic data
- GRE-level vocabulary in English MCQ — synonym/antonym questions are harder than any other BD bank exam

5. Three-Track Comparison — Difficulty & Scope Progression

The three examination levels are not simply 'harder versions' of each other — they test fundamentally different competency profiles. The AI platform must maintain three separate question banks, three difficulty calibrations, and three viva preparation tracks.

Dimension	Officer	Senior Officer	Assistant Director
Prelim Total	80 marks	100 marks	100 marks
Written Total	~110 marks	~145 marks	200 marks (7 sections)
Viva	15 marks	20 marks	25 marks
Math Complexity	SSC level (Percentage, Interest)	HSC level (Probability, Stats)	Advanced (Functions, Complex DI)
Bangla Depth	Basic grammar + literature overview	Advanced grammar + literary analysis	Literary criticism + language history
English Level	B2 equivalent	C1 equivalent	GRE / IELTS 7.0+ level
Banking Required	No (MCQ only)	Yes — operational banking	Yes — policy + regulatory depth
ICT Required	Basic (hardware, MS Office)	Mid (Networking, SQL, Cloud)	Advanced (OSI, DBMS, Fintech, Cyber)
Argument Writing	No	No	Yes — 30 marks (AD exclusive)
Viva Style	Personal + basic awareness	Operational banking Q&A	Policy-level + situational judgment
Typical Vacancy	500–1000+	1000–1500+	100–300 (most competitive)
AI Question Difficulty Tag	EASY	MEDIUM	HARD / ADVANCED

6. Written Exam Deep-Dive — All Levels

6.1 Focus Writing Topics — English & Bangla

Focus Writing is the highest-value written section. The AI must generate model essays, evaluate candidate submissions, and provide paragraph-level feedback. Topics are recurring across multiple BB exam years.

English Focus Writing Topics	Bangla Focus Writing Topics (বাংলা রচনা)
Inflation and economic stability	মূল্যস্ফীতি ও অর্থনৈতিক স্থিতিশীলতা
Artificial Intelligence and the future of work	কৃত্রিম বুদ্ধিমত্তা ও বাংলাদেশ
Climate change and sustainable development	জলবায়ু পরিবর্তন ও টেকসই উন্নয়ন
Sustainable Development Goals (SDGs)	টেকসই উন্নয়ন লক্ষ্যমাত্রা (SDGs)
Bangladesh Economy: Challenges and prospects	বাংলাদেশের অর্থনীতি: সম্ভাবনা ও চ্যালেঞ্জ
Financial Inclusion in Bangladesh	আর্থিক অন্তর্ভুক্তি
Digital Bangladesh and Smart Bangladesh Vision	ডিজিটাল বাংলাদেশ থেকে স্মার্ট বাংলাদেশ
Fourth Industrial Revolution	চতুর্থ শিল্প বিপ্লব
Green Banking and Environmental Finance	সবুজ অর্থায়ন ও পরিবেশ
Remittance and Bangladesh Economy	প্রবাসী আয় ও বাংলাদেশ
Gender equality and women empowerment	নারীর ক্ষমতায়ন ও সমঅধিকার
Digital financial services and MFS	মোবাইল ব্যাংকিং ও ডিজিটাল লেনদেন

6.2 Argument Writing (AD Exclusive — 30 Marks)

This section requires a structured for/against essay on a given statement. It is unique to the AD level and the most difficult written section. The AI platform needs a dedicated evaluation rubric.

Structure Component	Marks Weight	What Examiners Look For	Common Mistakes
Introduction + Thesis Statement	6 marks	Clear stance, context-setting, strong thesis	Vague opening; no clear position stated
Arguments FOR the statement	8 marks	3 distinct supporting points with evidence	Repetition; no factual support
Counterarguments + Refutation	8 marks	Acknowledging opposing view; refuting with logic	Ignoring counterarguments entirely
Conclusion	4 marks	Restates thesis; balanced final verdict	Copy-paste of introduction
Language & Structure	4 marks	Cohesive connectors; no grammar errors; varied vocabulary	Simple sentences; grammatical errors

6.3 Bangla Written Sections

Section	Format Requirements	AI Evaluation Criteria
বাংলা রচনা (Focus Writing)	ভূমিকা → মূল আলোচনা (৩-৫ অনুচ্ছেদ) → উপসংহার · ৫০০-৮০০ শব্দ	বিষয়বস্তু (15) + ভাষা শুদ্ধতা (10) + কাঠামো (10) = 35 marks
ভাবসম্প্রসারণ	উদ্ধৃতির আক্ষরিক অর্থ → প্রতীকী অর্থ → সামাজিক সত্য → উপসংহার	আক্ষরিক vs প্রতীকী বিশ্লেষণ · উদাহরণের প্রাসঙ্গিকতা
অনুবাদ (Bangla → English)	Register-appropriate translation · Idiomatic rendering · No literal word-for-word	Register accuracy · Idiom handling · False friend detection
বাংলা ব্যাকরণ	কারক, সমাস, সন্ধি, বাক্যশুদ্ধি, বানান, বাগধারা	Error-specific feedback per grammar rule violated

7. Viva Preparation Database

The viva database is the most complex knowledge base to build. BB viva panels include senior officials who ask layered, follow-up questions. The AI must simulate multi-level questioning chains, adapt to the candidate's academic background, and provide feedback on answer quality and delivery.

7.1 Viva Category Framework

Category	Sample Questions	Depth Required	Applies To
Personal & Motivation	Introduce yourself. Why Bangladesh Bank? Why leave your current job? Where do you see yourself in 5 years?	1-2 follow-up levels	All levels
Academic Background	(For ML Engineer) Explain gradient descent. What is overfitting? How does LLM work? What is a neural network?	3-5 follow-up levels	AD level
Basic Banking	What is a commercial bank? Difference between bank and NBF? What is CRR and SLR? What is the current repo rate?	2-3 follow-up levels	All levels
Advanced Banking	What is Basel III? Explain CAMELS. What is NPL classification in BD? What is Green Banking? Explain the money multiplier.	4-5 follow-up levels	AD + SO levels
Bangladesh Bank Specific	What are the functions of Bangladesh Bank? Name 5 BB departments. What is the BB Act 1972? What is the BRPD?	3-4 follow-up levels	AD level
Current Affairs	What is the current inflation rate? What is Bangladesh's GDP growth? What is the current FX reserve? Budget 2025-26 key figures?	2-3 follow-up levels	All levels
Monetary Policy	Explain the difference between repo and reverse repo. How does Bangladesh Bank control inflation? What is open market operation?	3-5 follow-up levels	AD + SO
Situational Judgment	A borrower defaults on a large loan — what steps does the bank take? A customer complaint escalates — how do you handle it?	2-3 levels	AD + SO levels
Stress / Pressure Questions	I don't agree with your answer. Are you sure? Can you prove that? The panel member was an examiner at your university.	1-2 levels — test composure	AD level
Salary & Joining	Are you ready to work in any branch? Can you start within 30 days? Are you	1 level	All levels

Category	Sample Questions	Depth Required	Applies To
	comfortable with government pay scale?		

7.2 Background-Specific Viva Tracks

Academic Background	Technical Question Topics	Recommended Viva Prep Focus
Engineering / CS / ML	ML concepts, algorithms, database design, networking, software development lifecycle, AI in banking	LLM, deep learning, system design, fintech applications, AI risk in banking
Economics / Finance	Macro/microeconomics, IS-LM model, Phillips curve, monetary theory, public finance, econometrics	Exchange rate models, growth theory, Keynesian vs. monetarist, BD fiscal policy
Commerce / Business / MBA	Financial accounting, management accounting, marketing, HRM, corporate governance	Ratio analysis, financial statements, banking operations, risk management
Science (Physics, Chemistry, Biology)	General science knowledge, scientific method, environmental science, climate	Link scientific knowledge to sustainability and green banking in BD
Arts / Social Science / Law	Bangladesh history, political science, governance, constitution, social development	Constitutional provisions, BB Act, legal framework of banking, AML law

7.3 Bangladesh Bank Institutional Knowledge — Viva Critical

- Bangladesh Bank Act 1972 — BB's mandate, governor appointment, board composition, monetary policy authority
- Bank Company Act 1991 — capital adequacy, single borrower exposure limit, board independence rules
- BB Departments: BRPD (Banking Regulation & Policy), FEPP (Foreign Exchange Policy), CMSD (Capital Market Supervision), HRD-1 (Human Resources), Research Department, SME & Special Programmes, DMFAS
- CRR (Cash Reserve Ratio) — current rate and its mechanism · SLR (Statutory Liquidity Ratio) — calculation and purpose
- Repo Rate & Reverse Repo Rate — current rates, how they transmit to commercial lending rates
- CAMELS rating framework — Capital Adequacy, Asset Quality, Management, Earnings, Liquidity, Sensitivity
- NPL classification — sub-standard, doubtful, bad/loss categories and provisioning requirements
- MFS regulation in BD — bKash, Nagad, Rocket; agent banking guidelines; e-KYC rules
- Green Banking and Sustainable Finance — BB's green taxonomy, ESG reporting, climate risk in banking
- Islamic Banking in BD — Islami Bank Bangladesh, Shariah Board, murabaha, musharaka, ijara, sukuk
- Basel III in Bangladesh — capital tiers (CET1, AT1, T2), leverage ratio, LCR, NSFR implementation timeline
- AML/CFT framework — BFIU (Bangladesh Financial Intelligence Unit), FATF compliance, STR/CTR reporting

8. Data Collection Strategy — AI Pipeline

The AI platform's quality depends entirely on the richness and accuracy of its training data. The pipeline must be structured across five tiers: primary question data, written content, viva intelligence, concept knowledge, and live economic data feeds.

Dataset Type	Primary Sources	What to Collect	Priority
MCQ (Primary)	PDF books: Saifur's, MP3, Oracle, Khairul's · Coaching notes	Question, 4 options, correct answer, full explanation, year, exam type, topic tag, difficulty level	CRITICAL
MCQ (Scraping)	jobstestbd.com · exam blogs · educational portals	Structured MCQs with topic mapping and correct answer verification	HIGH
MCQ (Video)	YouTube solution channels (Khairul's Math, Munzereen Shahid)	Repeated MCQs via Whisper transcription + explanation extraction	HIGH
MCQ (Social)	Facebook BB preparation groups · Telegram channels	Real repeated questions from recent exams (crowd-sourced verification required)	HIGH
Written Data	Coaching books · model test papers · student answer PDFs	Essays, translations, math solutions (with step-by-step), argument writing samples	CRITICAL
Written (AI-Generated)	Claude/GPT generation with expert review gate	Model answers for all Focus Writing topics at 3 quality levels (good/avg/poor)	HIGH
Viva Data	Facebook groups · YouTube viva interview discussions · Telegram coaching channels	Real candidate Q&A, follow-up chains, panel composition data, stress questions	HIGH
Viva (Structured)	Coaching center archives · One-on-one with past AD/SO candidates	Multi-level question chains (2–5 levels), background-specific Q banks	CRITICAL
Concept Data	Bangladesh Bank website · Bangladesh Economic Review · NCTB books · Wikipedia (filtered)	Definitions, frameworks, policy notes, banking concepts — with source citation	CRITICAL
Economy Data (Live)	Bangladesh Bank reports · BBS · IMF · World Bank · MoF budget	GDP, inflation, FX reserve, remittance, budget figures — updated quarterly	HIGH
Legal Data	BB Act 1972 · Bank Company Act 1991 · BRPD circulars · AML Act	Full text + structured summaries of key sections candidates need	HIGH

Dataset Type	Primary Sources	What to Collect	Priority
Past Papers	BB AD 2010–2025 · Senior Officer 2010–2025 · Officer 2010–2025 · Cash Officer	Full paper PDFs → structured MCQ + written questions with year tagging	CRITICAL

9. MCQ Data Sources — Structured View

Source Type	Platform / Resource	Value Level	Content Type	Collection Method
PDF Books	Saifur's Math / English · MP3 GK / Banking · Oracle Bangla / GK · Khairul's Bank Math	Very High	Structured MCQ — gold standard	Manual extraction + OCR
PDF Books	Professor's Bangla · Professor's GK · Master English · MP3 Bank Math	Very High	MCQ + explanations	OCR + parsing pipeline
Social Groups	Facebook: 'Bangladesh Bank Job Preparation' 'BB AD Preparation'	High	Real exam repeated questions	Manual collection + AI dedup
Messaging Apps	Telegram: BB exam prep channels · Coaching broadcast channels	Very High	Fast-updated recent questions	Bot scraping + verification
Web Sources	jobstestbd.com · campustimesbd.com · exam blogs · educatebd.com	Medium–High	Organised question archives	Structured scraping
Search Engine	Google: 'BB AD question PDF 2023 site:*' searches	High	Hidden past papers in PDF	PDF crawler + extraction
Video Transcription	YouTube: Khairul's Math, Munzereen Shahid, BankMath BD channels	High	Pattern repetition detection	Whisper ASR + timestamp extraction
Official BB Site	erecruitment.bb.org.bd · bb.org.bd/mediaroom	Critical	Official syllabus + notices + circulars	PDF scraper + structured parser

Data Quality Rules

Every MCQ must have: question text · 4 options · correct answer marked · source citation · year/exam tagged · subject → topic tagged.

Deduplication: semantic similarity check (cosine similarity > 0.85 = flag as near-duplicate). Keep one, tag variants.

Bangla content: human review gate mandatory before going live — AI Bangla accuracy is 70-80%, not production-ready without verification.

Statistics in GK and Banking questions: every number must have a date-stamped source. Outdated statistics must be flagged and updated quarterly.

Explanation requirement: every question needs a 2-5 sentence explanation of the correct answer — this is the primary learning asset.

10. Viva Data Model — Real Exam Behaviour

BB viva is not a flat Q&A session. Panels ask multi-level follow-up questions — if you answer the first question well, they go deeper. If you stumble, they probe harder. The data model must capture this dynamic chain structure.

Category	What to Collect	Follow-up Depth	Data Format
Personal	Introduction scripts · career motivation · background questions · 'why BB?' responses	1–2 levels	JSON: question → model_answer → follow_ups[]
Banking (Basic)	Repo rate, CRR/SLR, types of banks, central bank functions	2–3 levels	Question tree: root → L2 → L3
Banking (Advanced)	Basel III, CAMELS, NPL classification, monetary policy transmission	4–5 levels	Deep chain: root → definition → mechanism → BD-specific → current figure
BB Institutional	BB Act, departments, governor powers, circulars, BB organogram	3–4 levels	Structured fact sheet + Q chain
Current Affairs	Latest inflation rate, GDP growth, FX reserve, budget highlights	2–3 levels	Linked to live data feed — auto-updates quarterly
Technical (by background)	ML/DL for engineers · macro models for economists · accounting ratios for commerce grads	3–5 levels	Background-tagged question trees
Behavioral	Stress questions · panel disagreement simulation · rapid-fire rounds	1–2 levels (composure test)	Scenario cards with model responses
Follow-up Chains	2–5 level questioning chains from real BB viva candidate reports	Variable	Nested JSON: Q → A → follow_Q → follow_A

11. Concept Data — AI Knowledge Brain

The concept database is the semantic core of the platform — it powers explanations, viva AI responses, essay generation, and adaptive recommendations. Every concept node must link to the relevant exam sections where it appears.

Domain	Core Concepts	Linked Exam Sections
Banking	Monetary policy · Basel norms I–III · Financial systems · AML/KYC · Banking structure (BB + commercial) · CAMELS · NPL classification · BRPD circulars · Green banking · Islamic banking · MFS regulation · Agent banking	MCQ Banking (all levels) · Written GK · Viva all levels
Economics	Inflation (demand-pull, cost-push, stagflation) · GDP & national income accounting · Fiscal policy · Exchange rate systems · Development economics · Balance of payments · IS-LM model · Phillips curve · Money multiplier · Bangladesh Economic Review figures	MCQ GK & Banking · Written GK · Viva AD level
ICT	Networking layers (OSI model 7 layers) · TCP/IP protocol suite · Database concepts (RDBMS, SQL, normalization) · Cybersecurity (SSL/TLS, phishing, ransomware, 2FA) · Cloud computing (IaaS/PaaS/SaaS) · Fintech (blockchain, e-payment, MFS) · Digital transformation	MCQ ICT (all levels) · Viva AD (technical)
Bangla Language	Grammar rules: সন্ধি, সমাস, কারক, বিভক্তি, বাচ্য, উক্তি · Literature: রবীন্দ্রনাথ, নজরুল, মাইকেল, জীবনানন্দ, শরৎচন্দ্র · Language history: ভাষা আন্দোলন, বাংলা ভাষার উদ্ভব · Bangla writing rules: বানান (Bangla Academy standard)	MCQ Bangla · Written Bangla · ভাবসম্প্রসারণ
English	Grammar: Tense system, conditionals, clause types, articles, prepositions · Vocabulary: word roots, prefixes/suffixes, GRE word families · Writing: essay structure (introduction, body, conclusion), cohesive devices · Idioms and Phrases · Comprehension: inference, tone, main idea	MCQ English · Written comprehension + essay · Viva (English medium)
Mathematics	Formulas and shortcuts: percentage, SI/CI, P&L, ratio, work, distance · Advanced: probability rules, permutation-combination, arithmetic/geometric series, set theory, Venn diagrams · Written math: step-by-step solution presentation, partial marks strategy	MCQ Math · Written Math (30M)
Bangladesh Affairs	Liberation War 1971 (phases, key events, leaders, martyred intellectuals) · Constitution of Bangladesh (fundamental rights, state principles) · BD geography · Sectors and ministries · Awards and achievements · Famous personalities	MCQ GK · Written GK · Viva all levels

Domain	Core Concepts	Linked Exam Sections
Current Affairs	National budget key figures · GDP growth rate · Inflation rate (monthly BB data) · FX reserve · Remittance data · Global rankings (HDI, Ease of Doing Business) · Recent BB policies	MCQ GK · Written GK · Viva (current affairs) — updated quarterly

12. What Else to Add — Critical Missing Layers

Beyond the syllabus content, the AI platform needs several technical and structural layers that are not covered by subject knowledge alone. These determine whether the platform is a content library or a genuinely intelligent preparation tool.

12.1 Platform Intelligence Features — Not in Original Plan

Feature	Priority	Description	Missing From
Negative Marking Strategy Engine	CRITICAL	Given a candidate's accuracy % per subject, calculates optimal number of questions to attempt in the 100Q prelim. Simulates expected score under different attempt counts.	Original plan
Full Mock Exam Simulator	CRITICAL	Exactly replicates prelim (100Q, 1 hour, -0.25 per wrong, locked sections) and written (200M, 2 hours, 7 sections with suggested time budgets) conditions.	Original plan
Argument Writing Evaluator	CRITICAL	AI rubric for the 30-mark AD Argument Writing section: thesis (6) + for-arguments (8) + counterargument/refutation (8) + conclusion (4) + language (4).	Original plan
Written Math Partial-Marks Trainer	CRITICAL	Teaches step-by-step solution formatting for the 30-mark written math section. Partial marks awarded per step — showing working is as important as the answer.	Original plan
Comprehension Answer Format Trainer	HIGH	Teaches the BB-specific comprehension answer format. Most candidates know the answer but lose marks by not following the expected reference-to-text structure.	Original plan
Precis Writing Module	HIGH	Strict Precis rules: must be 1/3 of original length, no first person, indirect speech, specific title format. AI evaluates all these mechanically before language quality.	Partial mention only
Current Affairs Auto-Pipeline	HIGH	Weekly digest auto-generated from bb.org.bd circulars, Bangladesh Economic Review, BBS data, IMF/World Bank BD updates. Formatted into MCQ-ready GK facts.	Original plan
Candidate Profile & Track Selector	HIGH	Collects: target post (AD/SO/Officer), academic background, available hours/day, exam date, exam history (1st attempt / repeat). Generates personalised study plan and viva track.	Original plan
ভাবসম্প্রসারণ AI Scorer	HIGH	Evaluates the Bangla concept expansion format: literal meaning → symbolic meaning → general social truth → conclusion. Each layer scored separately.	Mentioned, not detailed
Document Verification Checklist	MED	Exact document order BB requires for viva day. Common rejection reasons. How to present the educational journey coherently to the document checking board.	Original plan
Post-Mock Analytics	MED	After every mock: time per question, accuracy by section, negative marking impact, effective score vs. raw score, cohort percentile rank.	Original plan



Feature	Priority	Description	Missing From
Stress Viva Simulator	MED	Panel AI disagrees with answer, raises aggressive follow-ups, creates time pressure. Tests composure, not just knowledge.	Original plan

13. Recommended Books & Reading Materials

Subject	Books (Primary)	Additional Resources	Used For
Mathematics	Khairul's Bank Math · Saifur's Math · MP3 Bank Math	BCS Math by Professor's · YouTube: Khairul's Basic Math channel	MCQ + Written Math
English	Saifur's English · Master English · Cliffs TOEFL	Munzereen Shahid YouTube · BBC Learning English · IELTS prep materials	MCQ + Focus Writing + Comprehension
Bangla	Professor's Bangla · Oracle Bangla	Bangla Academy Grammar (official) · BD Shilpakala Academy literature notes	MCQ + Written Bangla
General Knowledge	MP3 GK · Professor's GK · Oracle GK	Daily newspaper (Prothom Alo, Daily Star) · Current Affairs digest monthly	MCQ + Written GK + Viva current affairs
Banking & Economics	Bangladesh Economy by Abul Barakat · Banking Diploma Foundation Materials	Bangladesh Economic Review (MoF, annual) · BB Annual Report · IMF Bangladesh reports	MCQ Banking · Written GK · Viva
ICT	ICT section: Khairul's or Saifur's combined book · NCTB ICT text (HSC level)	YouTube: Networking tutorials · Cisco CCNA basics · BB FinTech circular documents	MCQ ICT · Viva (AD technical)
Bangladesh Bank Specific	BB Act 1972 (full text) · Bank Company Act 1991	BRPD circulars (bb.org.bd) · BB Annual Report 2023-2024 · Bangladesh Bank organogram	Viva preparation (AD level)
Argument Writing	No dedicated BD book — use IELTS Writing Task 2 practice books	IELTS Cambridge Practice Tests · Saifur's Writing (partial) · TOEFL Integrated Writing	Written Exam — AD Argument Writing

14. Implementation Roadmap — Knowledge Base Build Sequence

The knowledge bases must be built in a specific sequence — each layer unlocks the next. Starting with the MCQ question bank is the correct first step because it establishes the subject taxonomy that all other features depend on.

Phase	Knowledge Base Task	Output	Unlocks	Timeline
Phase 1	Subject taxonomy build	BPSC syllabus → structured Subject > Chapter > Topic hierarchy for all 3 levels	All feature tagging	Month 1
Phase 1	MCQ bank seed (Officer level)	5,000+ verified MCQs (Bangla, English, Math, GK, ICT, Reasoning) with explanations	Adaptive quiz engine	Month 1–2
Phase 1	Concept data — Banking & Economics	500+ concept definitions with BB-specific examples, linked to exam sections	MCQ explanations + viva AI	Month 1–2
Phase 2	MCQ bank expand (SO + AD level)	10,000+ total MCQs across all 3 difficulty tracks; Banking & Finance MCQs added	Three-track quiz engine	Month 2–3
Phase 2	Written model answers	30+ Focus Writing model essays (Bangla + English) · 20+ Math solutions with step format · Argument Writing samples	Written evaluation AI	Month 2–3
Phase 2	Viva Q&A database (basic)	500+ Q&A pairs with 2–3 level follow-up chains; personal + basic banking categories	Viva simulator (basic)	Month 3–4
Phase 3	Current affairs pipeline	Weekly auto-digest from BB circulars + BBS + MoF data → formatted GK facts	GK questions + viva current affairs	Month 3 (ongoing)
Phase 3	Advanced viva Q bank (AD level)	200+ deep banking + institutional + background-specific Q chains	Viva simulator (advanced)	Month 4–5
Phase 3	Past paper complete archive	BB AD 2010–2025 + SO + Officer — all structured with year/level tagging	Question frequency analysis	Month 4–5
Phase 4	Content quality validation	All Bangla content: human review gate. All statistics: source-cited. All math: computationally verified.	Production readiness	Month 5–6
Phase 4	Negative marking + mock simulators	Platform intelligence layer — strategy engine, full mock exam, post-mock analytics	Platform differentiation	Month 5–6

Implementation Priority: The Three Non-Negotiables

1. Human review gate for ALL Bangla AI-generated content — deploy without this and wrong grammar will destroy user trust on day one.
2. Source-cited statistics for all Banking/Economics content — BB officials know the exact current figures; your platform must too.
3. Argument Writing evaluator live before launch — it's 30 marks and no competitor has it; shipping without it wastes the biggest differentiation opportunity.

Manual_data_collection:

We will collect data by scraping publicly available books and educational resources, then store the raw data in a centralized database. In the initial processing phase, the data will be cleaned and normalized by removing duplicates and standardizing categories.

In the second phase, the dataset will be continuously enriched with daily updates from newspapers, current affairs sources, and verified news outlets to ensure the system remains up-to-date and contextually relevant.